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How elephants avoid cancer

Pachyderms have extra copies of a key tumour-fighting gene.

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Theo Allofs/Minden Pictures/FLPA

Multiple copies of a tumour-suppressor gene help elephants avoid cancer.

Elephants have evolved extra copies of a gene that fights tumour cells, according to two independent studies^{1, 2} — offering an explanation for why the animals so rarely develop cancer.

Why elephants do not get cancer is a famous conundrum that was posed — in a different form — by epidemiologist Richard Peto of the University of Oxford, UK, in the 1970s³. Peto noted that, in general, there is little relationship between cancer rates and the body size or age of animals. That is surprising: the cells of large-bodied or older animals should have divided many more times than those of smaller or younger ones, so should possess more random mutations predisposing them to cancer. Peto speculated that there might be an intrinsic biological mechanism that protects cells from cancer as they age and expand.

At least one solution to Peto's paradox may now have been found, according to a pair of papers independently published this week. Elephants have 20 copies of a gene called *p53* (or, more properly, *TP53*), in their genome, where humans and other mammals have only one. The gene is known as a tumour suppressor, and it snaps to action when cells suffer DNA damage, churning out copies of its associated p53 protein and either repairing the damage or killing off the cell.

The elephant's tale

Uncovering *TP53*'s role has taken a few years. Joshua Schiffman, a paediatric oncologist and scientist at the University of Utah in Salt Lake City, first heard about Peto's paradox three years ago at an evolution conference, when Carlo Maley, an evolutionary biologist now at Arizona State University in Tempe, revealed he had found multiple copies of *TP53* in the African elephant's genome.

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Schiffman specializes in treating children missing one of their *TP53* gene's two alleles, which leads them to develop cancer. So after hearing Maley's talk, he wondered whether elephants held some biological insight that could help his patients. He teamed up with Maley, who had not yet published his work, and asked elephant keepers at Salt Lake City's zoo whether they could spare some elephant blood so that he could test how the p53 protein works in the mammals' white blood cells.

At about the same time, in mid-2012, Vincent Lynch, an evolutionary geneticist at the University of Chicago in Illinois, was preparing for a lecture on Peto's paradox, and wondered about mechanisms that could explain it. "Right before I gave the lecture, I searched the elephant genome for *p53*, and 20 hits came up," says Lynch.

Schiffman and Lynch's teams have now independently revealed their findings — Schiffman's in the *Journal of the American Medical Association*¹, and Lynch's in a paper² posted to the bioRxiv.org preprint site, but which is in review at the journal *eLife*.

Using zoo autopsy records for 36 mammals — from striped grass mice to elephants — Schiffman's team recorded no relationship between body size and cancer rate. (Around 3% of elephants get cancer, according to the team's analysis of hundreds of captive-elephant deaths).

The researchers found that elephants produce extra copies of the p53 protein, and that elephant blood cells seem exquisitely sensitive to DNA damage from ionizing radiation. The animals' cells carry out a controlled self-destruction called apoptosis in response to DNA damage at much higher rates than do human cells. Schiffman suggests that, instead of repairing the DNA damage, compromised elephant cells have evolved to kill themselves to nip nascent tumours in the bud.

"This is a brilliant solution to Peto's paradox," he says.

Mammoth set

Lynch's team — working with African and Asian elephant skin cells from the San Diego Zoo in California — found similar results. They also discovered more than a dozen *TP53* copies in two extinct species of mammoth, but just one copy in elephants' close living relatives, manatees and hyraxes (a small, furry mammal). Lynch thinks that the extra copies evolved as the lineage that led to elephants expanded in size. But he thinks that other biological mechanisms are involved too.

Mel Greaves, a cancer biologist at the Institute for Cancer Research in London, agrees that *TP53* cannot be the only explanation. "As large animals get bigger, they become more and more sluggish," he notes, thereby slowing their metabolism and the pace at which their cells divide. And protective mechanisms can only do so much to stop cancer, he adds. "What would happen if elephants smoked and had a bad diet," he says. "Would they really be protected from cancer? I doubt it."

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Majid Ali • 2015-12-18 05:58 AM

What Might Elephants Say About Human Cancer? One in five humans die of cancer. Only one in 33 elephants get cancer, according to a recent analysis of hundreds of captive-elephant deaths. The blind mole rat is also a remarkably cancer resistant rodent. The size does not seem to matter here. Colon cancer is the most common cancer in humans. It nearly always arises at the surface of colonic polyps. Among such polyps, malignant transformation occurs more commonly in large, sessile villous adenomas than in smaller, adenomatous polyps with a smaller population of neoplastic cells. The same holds for endometrial hyperplasia, which progresses from its mild to moderate to severe forms with incremental cellularity, and evolving into cancer with yet larger population of cells. The size and cellularity seem to matter here. Humans have one copy of cancer-suppressive P53 genes. Elephants have 20 copies of the gene. So, the number of copies of this gene seems to matter. Blind mole rat has a mutated P53 only gene. So, the number of copies of this gene does not seem to tell the whole story. First identified in 1979, p53 was voted the molecule of the year by Science magazine in 1983. Notwithstanding large strides in our understanding of carcinogenesis, to date the P53 field of inquiry has not met the early expectation in the control of cancer. In a desert, what has primacy, the soil or the seed? In 1980, after a study of over 1,000 colorectal biopsies as a hospital pathologist, I recognized the need for ecologic thinking in clinical medicine and published a monograph entitled *Altered States of Bowel Ecology* (ref 1) to focus on the matter of the oral-stomach-bowel flora and the centrality of the bowel in all deliberations of health/dis-ease/disease continuum. I recognized the profound ecologic disruption caused by heavy sugar intake, frequent use of antibiotic abuse, neglect of mold and food allergy, and chronic stress. In the monograph, based on personal clinical experience with patients with chronic immune-inflammatory disorders, I described my seed-feed-and-weed guidelines for restoring gut ecology in such patients. In 2004, in the journal *Townsend Letter*, I published evidence of Krebs cycle disruption in 236 patients with immune-inflammatory disorders (ref 2). In 2015, *Nature* published online the data obtained with a followup study of 351 patients. (ref 3). In 2010, *Nature* addressed this with the following words: "By 2020, personalized health care could involve doctors monitoring the metabolic activities of a patient's gut microbes and, possibly, modulating them therapeutically" (*Nature* 2010;463:32). Next, consider the following words, again from *Nature* (2014;516:302): "World Health Organization warns that world may be heading into a 'post-antibiotics era'" I return to the question, what might elephants say about cancer? Elephants, of course, have had an advantage over humans. Until recently, the gut ecosystems of these vegetarians have not been violated by incremental quantities of environmental carcinogens. Human bowels have not been so fortunate. Might elephants say the following: in understanding and treating malignant tumors, cancer genetics are important and equally important are cancer epigenetics, most importantly the matters of ecology of the alimentary tract. Humans should learn a thing or two about environmental carcinogens. They

might add: our herds used to ramble along in peace and that helped as well. Alas, all that is changing. References: 1. Ali M. Altered States of Bowel Ecology. Teaneck, NJ, 1980. 2. Ali M. Respiratory-to-Fermentative (RTF) Shift in ATP Production in Chronic Energy Deficit States. Townsend Letter for Doctors and Patients. 253: 64-65 (2004). 3. Ali M. Succinate retention. In: Chouchani ET, Pell VR, Guade E, et. al. Ischaemic accumulation of succinate controls reperfusion injury through mitochondrial ROS. Nature 515, 431–435 (2014).



David Gurwitz • 2015-10-11 09:47 AM

This is an amazing discovery that hopefully will improve research into healthy aging. Further clues to healthy aging (and not related to the p53 pathway) come from the recently published DNA sequence of the bowhead whale, probably having the longest life span of all creatures: Keane M et al. Insights into the evolution of longevity from the bowhead whale genome. Cell Rep. 2015 Jan 6;10(1):112-22.

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4536333/> David Gurwitz Tel Aviv University Faculty of Medicine

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